

is obviously no original picture of a dragon or a unicorn the child has ever possibly seen. But he or she has gone through drawings and artist impressions, which is a form of representation from imagination. It is data about something that does not exist. The data is entirely made up to help the child identify the dragon even if it never existed. Similarly, even for computers, in such cases, the data has to be made up. Made-up data is also known as ‘synthetic data’ and plays a very important part in the training of machine learning models.

But this is just about object recognition. Machine learning models can be used for a lot more purposes than just object recognition. For example, a model can be made to identify fraudulent transactions from bank statements. However, there may not be so many records of fraudulent transactions available to train the model. In such scenarios, synthetic data has to be prepared.

A very different type of artificial neural network called generative adversarial neural (GAN) network is used to generate synthetic data. But what are the moral obligations to the generation and use of synthetic data? That boils down to the fact that synthetic data is not real data and does not have a physical significance. For example, let’s look at this question — how safe is lying? Well, telling a child that a dragon lives in the mountains may be safe, but stating someone was killed by a dragon is not legal and an insurance fraud. It is the same lie, but the situation it is used in shows whether it is safe and legal.

As we approach the end of 2023, recent months have witnessed a substantial increase in the capabilities of AI. Generative AI now crafts stories, draws pictures, and performs various tasks with heightened proficiency. But GANs have been around for a long time. It’s just that generative AI has made them more accessible to everyone. This has increased cybercrimes and frauds a



Figure 2: Picture created by DALL-E-3 with the prompt ‘Realistic image of a teenager fighting a dragon in mountains, cloudy sky, teenager is using a sword, dragon breathing fire.’

lot. For example, the frauds over voice calls sound more realistic now because the scammer can actually use AI to modulate his or her voice to the voice of someone the victim knows. For the past few months, after Photoshop added the Generative AI fill feature, it has been a joke that people could fake being sick by photoshopping being admitted into a hospital. This shows how easy it is to tamper with evidence these days.

A prime example is deepfakes, which can accurately swap a person’s face with another with a high level of accuracy, often escaping detection by the naked eye. A recent example might be the deepfake of a popular south Indian actress doing the rounds on social

media, which was made from a video of a social media influencer. Deepfakes have been used by many criminals and in many cyber frauds. There has been a rising case of deepfakes on indecent pictures or videos to harm someone’s reputation.

With the amount of synthetic data being pumped into society every single day, it is quite difficult to differentiate between reality, the illusion, and the deception and lies. As children, we trusted what we were told, and believed in dragons, unicorns, and fairies. But we soon outgrew that. As adults we may now need to enhance our discerning capabilities substantially to figure out what’s real and what’s not. **END** 🐼

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